To-Do List Application

Python Mini Project

A simple desktop app to add, finish and delete daily tasks, and save them in a file.

Name and Enrollment No.

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Department

IT-A

Subject

Python Programming

About the Project

A small desktop app to manage daily tasks.

- Add a new task with one click or Enter
- Mark a task as done or undo it
- Delete one task or clear all finished tasks
- Tasks are saved in a file and reload on start

! Built for learning: lists, functions, file handling and a simple GUI.

4 task operations

1 Python file

1 JSON data file

Objectives of the Project

1 Use Lists and Dictionaries

Store every task as a dictionary inside a list.

2 Write Functions

Keep loading, saving and each action in small functions.

3 Save Data in a File

Keep tasks in tasks.json so they stay after closing.

4 Build a Simple GUI

Make a clean window with Tkinter that is easy to use.

Concepts and Technologies Used

1 Python 3

Main language of the project

2 Lists and Dictionaries

Each task is a dictionary inside a list

3 Functions

load_tasks(), save_tasks() and one method per action

4 File Handling

Reads and writes tasks.json

5 Exception Handling

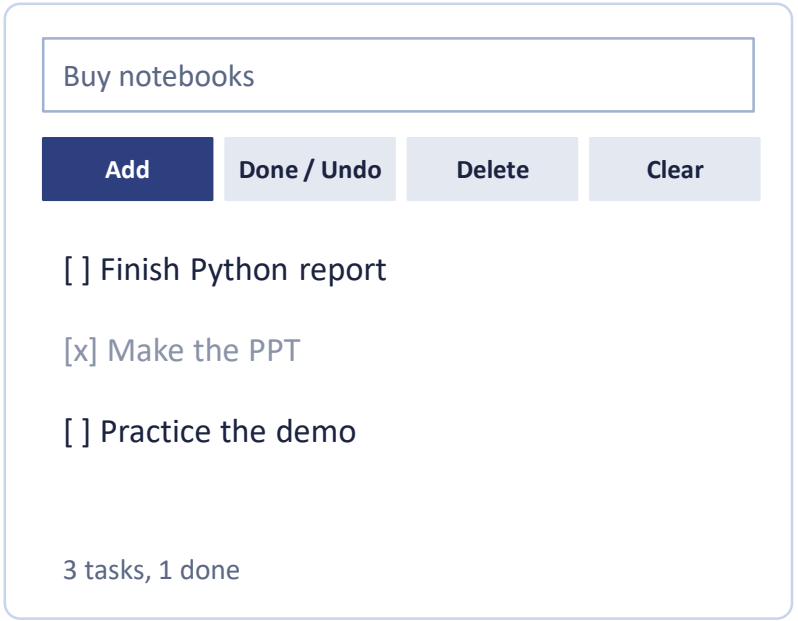
try/except for missing or broken files

6 Tkinter

Window, buttons, entry box and list

Main Features

1. **Add Task**
   Type a task, press Add or Enter
2. **Done / Undo**
   Mark the selected task as finished
3. **Delete Task**
   Remove the selected task
4. **Clear Done**
   Remove all finished tasks at once
5. **Auto Save**
   Every change is saved to tasks.json



Project Structure and Working

```
To-Do-List/
├── todo_app.py
│     ├── load_tasks()
│     ├── save_tasks()
│     └── class TodoApp
└── tasks.json
```

1. Start	load_tasks() reads tasks.json into a list
2. Act	Add, mark done or delete changes the list
3. Show	refresh() redraws the task list on screen
4. Save	save_tasks() writes the list back to the file

Key Code: File Handling

```
def load_tasks():
    try:
        with open(DATA_FILE) as f:
            return json.load(f)
    except (FileNotFoundError, json.JSONDecodeError):
        return []

def save_tasks(tasks):
    with open(DATA_FILE, "w") as f:
        json.dump(tasks, f, indent=2)
```

Reads safely

A missing or broken file gives an empty list, so the app never crashes

Stores clearly

Each task is saved as a title and a done flag in JSON

Conclusion and Future Scope

Conclusion

The To-Do List Application is a simple and useful Python project. It uses lists, functions, file handling and Tkinter together, and it keeps the user's tasks safe between runs.

Future Scope

1. **Due dates and reminders**
2. **Priority levels and colors**
3. **Search and filter tasks**
4. **Cloud sync or a mobile app**

Thank You

Any questions?

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